CCC-C-426D
August 12, 1970
SUPERSEDING
Fed. Spec. CCC-C-426C
January 3, 1966

FEDERAL SPECIFICATION

CLOTH, DRILL, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton drill cloth.

1.2 Classification.

1.2.1 Types and classes. The cloth shall be of the types as specified in table I and in the following classes as specified (see 6.2):

Class 1 - Greige.

Class 2 - Desized, preshrunk.

Class 3 - White or dyed, preshrunk.

NOTE: The greige texture for each type is that designated as class 1 in table I.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276 - Thread, Cotton.
PPP-P-1134 - Packaging And Packing Of Cotton And Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

FSC 8305

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Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.

Fed. Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

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3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton, spun into singles yarn for both warp and filling.

3.3 Color. Classes 1 and 2 cloths shall be natural color. Class 3 cloth shall be bleached or dyed as specified (see 6.2). The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.4.

3.3.1 Matching. The dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness. The dyed and finished cloth shall show colorfastness to laundering, wet drycleaning, perspiration, light and crocking equal to or better than the standard sample. When no standard sample is established or when the standard sample is not a reference for colorfastness, the dyed and finished cloth shall show "good" colorfastness to laundering, wet drycleaning, perspiration and light and shall show a Munsell Value for crocking not lower than 8.5. Testing shall be as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

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TABLE I. Physical requirements

Type	Class	Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength pounds (minimum)		Weave (for all types)
		Ounces	Warp	Filling	Warp	Filling	
I	1	7.0	72	60	115	82	3 harness 2/1 left hand twill
	2	7.0	74	58	105	75	
	3	6.5	74	56	105	70	
II	1	7.0	72	48	115	77	
	2	7.0	74	46	105	63	
	3	6.5	74	44	105	62	
III	1	6.7	72	48	110	65	
	2	6.5	74	46	100	60	
	3	6.0	74	44	90	55	

3.4.1 Width. The width shall be as specified (see 6.2), and shall be the minimum acceptable width inclusive of selvages.

3.5 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 12.0 percent for class 1, 5.0 percent for class 2, or 2.0 percent for class 3 when tested as specified in 4.4.

3.6 pH. The pH value of the water extract of the dyed and finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.7 Shrinkage. Classes 2 and 3 cloths shall be preshrunk so that the residual shrinkage shall not exceed 2.0 percent in either the warp or filling direction when tested as specified in 4.4. The preshrinking process used shall not be identified by name or trademark on the cloth, ticket or package.

3.8 Seam efficiency. The cloth for all types and classes shall have a seam efficiency of not less than 80 percent when tested as specified in 4.4.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces as specified in PPP-P-1134.

3.10 Fiber identification. Each piece shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

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3.11 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Defects found during examination shall be classified in accordance with 4.2.2.1 through 4.2.2.4. In addition, the sample shall be examined for the required basic color. The lot shall be unacceptable if the required color does not conform.

4.2.2.1 Yard-by-yard examination. Each piece in the sample shall be examined on the face side only. When the total yardage in the piece does not exceed 100 yards, the entire yardage in the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1, except that only those slubs and knots which exceed the limits shown on figure 1 of Fed. Std. No. 4 shall be scored.

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No folded linear yard shall be penalized more than 4 points. The sample size shall be 20 pieces selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds the following point values:

Class 1 - 45 points.
Class 2 - 45 points.
Class 3 - 35 points for dyed cloth.
42 points for bleached cloth.

The lot shall be unacceptable if the points per 100 square yards of two or more individual pieces exceeds the following point values:

Class 1 - 68 points.
Class 2 - 68 points.
Class 3 - 53 points for dyed cloth.
63 points for bleached cloth.

If one piece exceeds the point level, a second sample of 20 pieces shall be examined only for individual piece quality examination. The lot shall be unacceptable if one or more pieces in the second sample exceeds the point level. For individual piece quality, examination of local defects only shall be scored. Overall type defects shall be scored only in determining compliance with the lot average quality requirements. Point computation for lot quality and individual piece quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension - one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension - two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension - three points
For defects exceeding 9 inches in any dimension - four points

The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy (when applicable).
Excessive neppiness.

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4.2.2.2 Examination for length.

4.2.2.2.1 Individual pieces. During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for identification of preshrinking process and compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more bolts in the sample contain identification of a preshrinking process by name or trademark on the cloth or ticket, or are not labeled, ticketed, or invoiced in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine whether packaging, packing and marking comply with the section 5 requirements of this specification.

4.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable and as listed in table III shall be followed. The physical and chemical values specified in section 3, except when otherwise indicated, apply to the results of the determination made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 2-1/2 yards full width of the class 1 cloth, and 4-1/2 yards full width of the classes 2 and 3 cloths. The lot size shall be expressed in units of 1 yard. The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Test methods

Characteristics	Requirement paragraph	Test method
Yarn identification	3.2.1	1200 1/
Presence of labile sulfur	3.3	2020
Colorfastness to:		
Laundering	3.3.2	5610
Wet drycleaning	3.3.2	5622
Perspiration	3.3.2	5680
Light	3.3.2	5660
Crocking	3.3.2	5651
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4	Visual 2/
Nonfibrous material	3.5	2611
pH	3.6	2811
Shrinkage	3.7	5550
Seam efficiency	3.8	5110 3/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and the results reported as "pass" or "fail".

3/ The needle thread shall be ticket No. 24, 3 ply and the looper thread shall be ticket No. 60, 3 ply conforming to type IA1 of V-T-276. The needle shall measure 0.049 ± 0.001 across the blade at the top.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2)

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in clothing and equipage items.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Color required (see 3.3).
(d) Width required (see 3.4.1).
(e) Length if other than specified (see 3.9).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Air Force - 82

User activities:

Navy - MC, YD

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

CCC-C-426D; Cloth, Drill, Cotton

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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